

Practical Applications of LLMs

CSCC01 - Introduction to Software Engineering

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About Me



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Product Innovation in Population Health AI

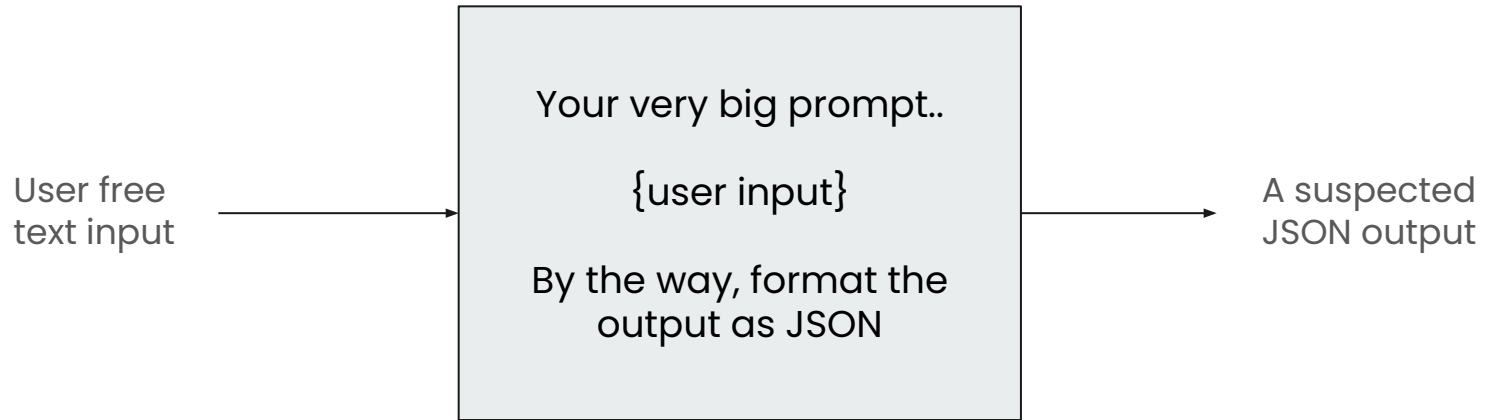
Sessional Lecturer II @ UTSC

CSCD01 – Engineering Large Software Systems

CSCC09 – Programming on the Web

Old LLM Applications – chatbots

Complete reliance on underlying LLM understanding,
low degree of control.



1 Big Prompt

Pros

- Simple
- The better the LLM, the better the results

Cons

- Control only possible through “prompt engineering”
- The worse the LLM, the worse the results

Workflows

Structured, deterministic sequence of steps designed for repeatable tasks.

Example: Resume Building

1. Obtain the job description / your job experience
2. Match what the job requires to your experience
3. Rephrase your experience to match what the job description requires

Sequential Workflow Demo

With [Mastra](#)

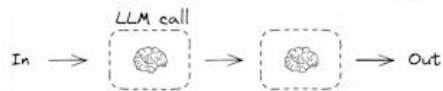
Why use an AI framework?

Other popular AI frameworks:

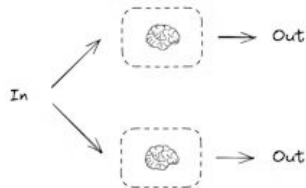
- Langgraph
- Google ADK

More workflows

Prompt Chaining

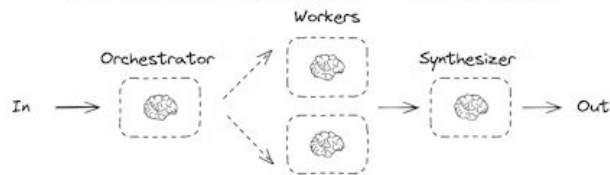


Parallelization

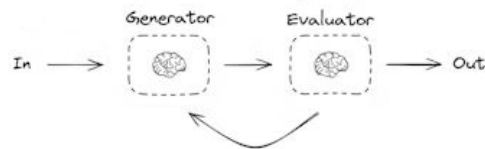


LLM is embedded in
predefined code paths

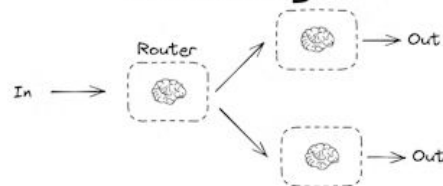
Orchestrator-Worker



Evaluator-optimizer



Routing



LLM directs control flow
through predefined code paths

Problem

Fully autonomous.

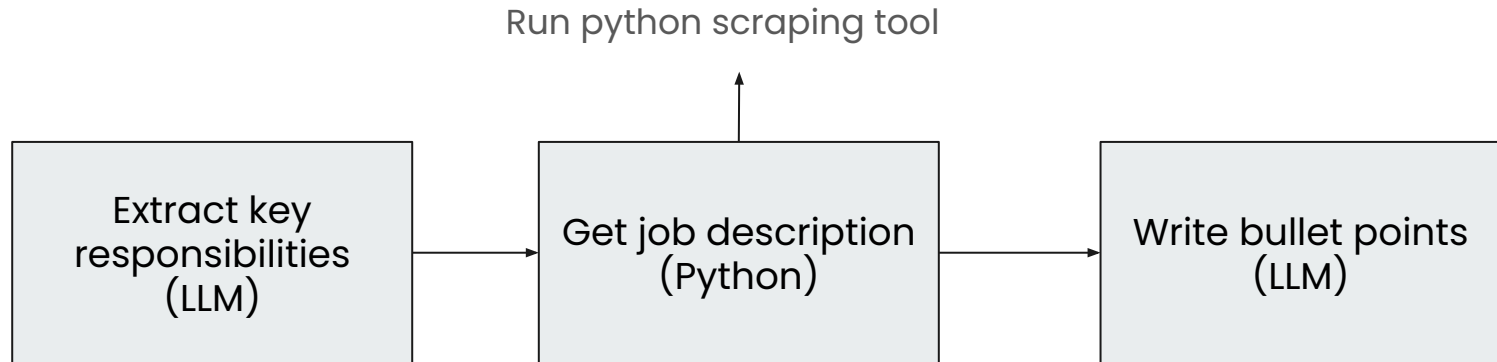
Gives you a good first draft only.

Quality depends entirely on the LLM.

Maybe it's a 70% match to what you really need.

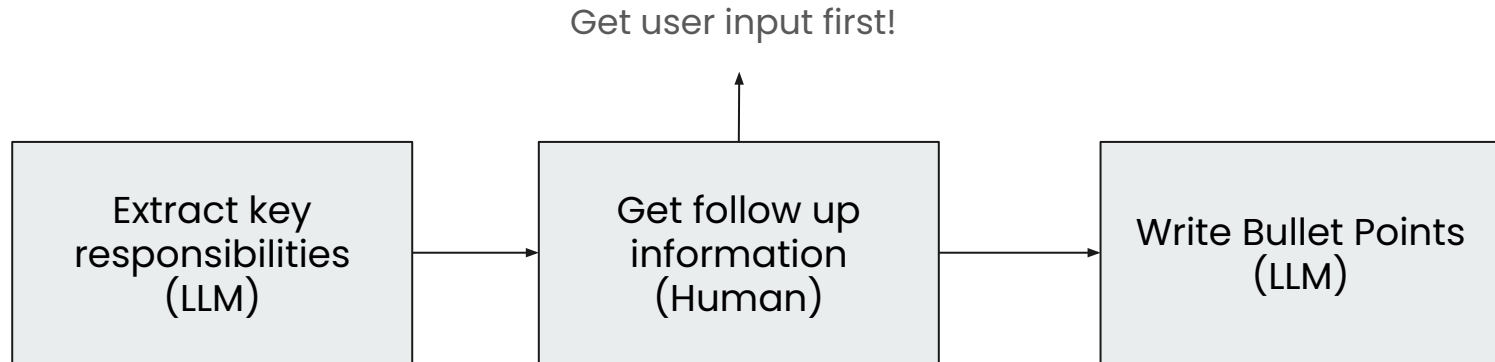
Function calling workflow

Add deterministic steps to ensure workflow reliability.



Human-in-the-loop workflow

Ask AI to discover what is missing, and prompt the user for follow-up information.

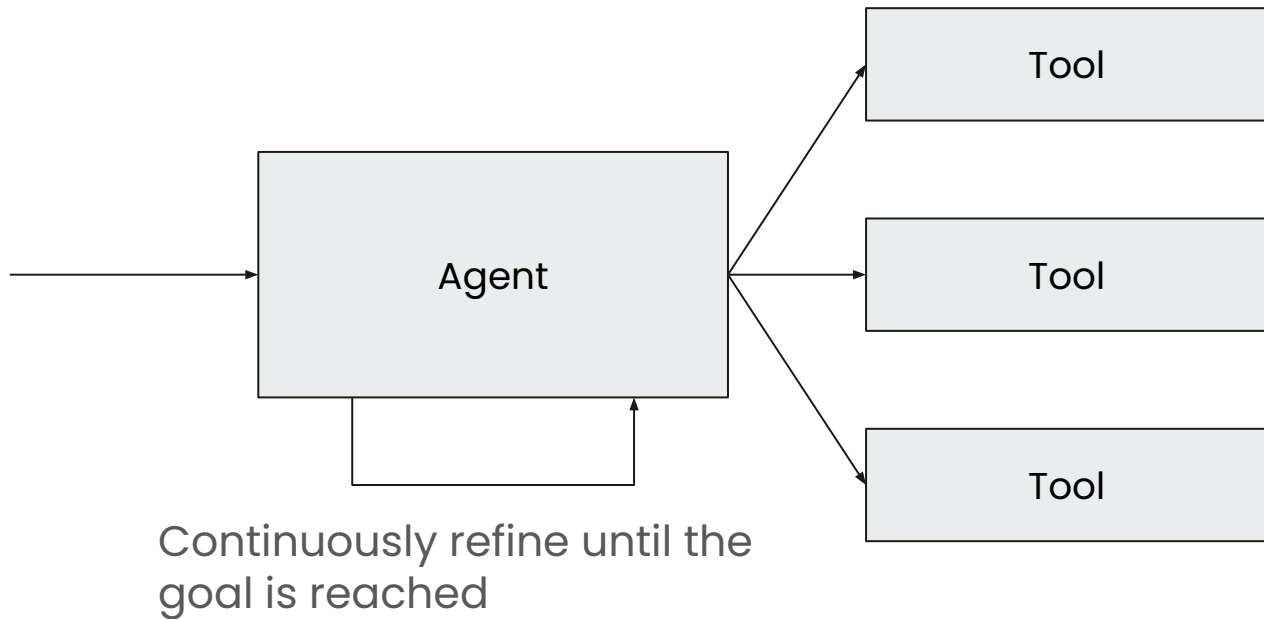


Workflows are limiting – it can do predefined things

What if we want to do more open ended tasks? It is impossible to know what users will do in advance

Agents

AI-driven process that uses reasoning to dynamically **decide on steps** to take and the tools to use to solve more open ended problems.



Tools

Use tools when an agent needs additional context more deterministically.

Similar to function calling – predetermined inputs and outputs used by the LLM. The outputs are used as LLM prompt input.

Agent Demo

With [Mastra](#)

- Barebones Agent (useless)
- Tool calling agent
- Workflow as tool calling agent
 - Human in the loop
- What is it doing under the hood? (Observability)

Reusable Tools

What if you want your agent to interact with external 3rd party integrations?

View JIRA ticket -> Code -> Create Pull Request

MCP Servers – remote tools

Model Context Protocol servers – accessed via HTTP endpoints.

Interacted in the same way as tools.

Great examples of Agents

- Claude Code
- Github Copilot
- Cursor

What tools does it have access to?

Trade-offs

How does the LLM know that a tool is available?

Context window

You have limited tokens per input prompt to LLM.

An input prompt can contain:

- System prompt (you write that)
- User input (could be long)
- Tool definitions
- Tool output
- etc.

What if you want your agent to access 100+ tools?

Context Engineering

Control the size of your prompt to only have relevant information.

[Context Engineering Guide](#)

[Cloudflare: Agent code mode](#)

In Summary

Workflow – well-defined tasks

Agents – open-ended questions enhanced by well-defined tasks

2 interventions:

- Human in the loop
- Tool calling

Quality Assurance

Testing software

Defined inputs

Expected output

Actual output

How does this apply to LLM outputs?

Evaluating LLM outputs

User inputs are not fixed, intent may come in different forms.

Even with same inputs, outputs likely is different.

Scoring

- Real human feedback
 - Manual labeling
- LLM-as-a-judge

LLM-as-a-judge

- Accuracy and reliability
 - Faithfulness to prompt
 - completeness
- Output quality
 - Toxicity
 - Keyword coverage
 - bias

Proactive Scoring & refinement

1. LLMs to generate synthetic inputs
2. Run synthetic inputs through pipeline
3. Evaluate outputs
4. Refine prompts and restart step 2

Likely, this happens in CI when developing the agent.
Scoring must pass threshold.

Reactive Scoring

1. Receive real user input
2. Run pipeline
3. Run scoring and save for further action

Demo!

Q&A

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Let's connect.



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